Audited m=51 conductor-271 volcano collision-walk record

Date: 2026-10-13

Status: **valid for the frozen scoped inference.** The sealed analyzer accepted all 400 jobs and all 25,600 verified planted-key solve records. All five held-out A/A gates passed. This is a bounded source-operation experiment, not a transfer experiment or a curve-security result.

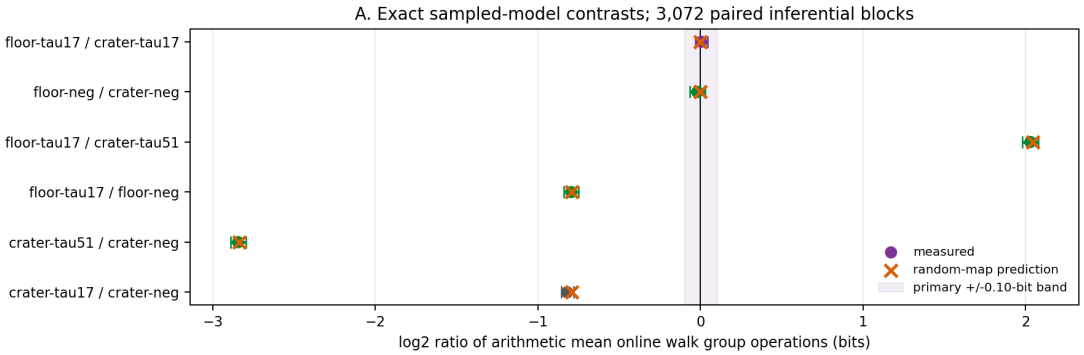
Outcome

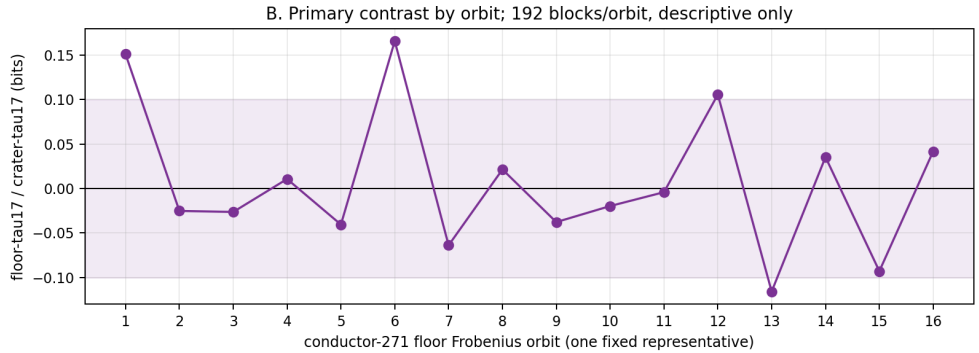
The primary same-action comparison was `floor-tau17 / crater-tau17 = +0.005763` walk-operation bit, with fixed-sample 95% CI `[-0.032170, +0.043374]`. The interval includes zero and is wholly inside the predeclared `+/-0.10`-bit practical-equivalence band. The permitted conclusion is therefore narrow: **no same-action sampled-level/model association larger than 0.10 collision-work bit was resolved for this implementation, subgroup, and exact sampled models.** It is not an estimate of a causal conductor effect.

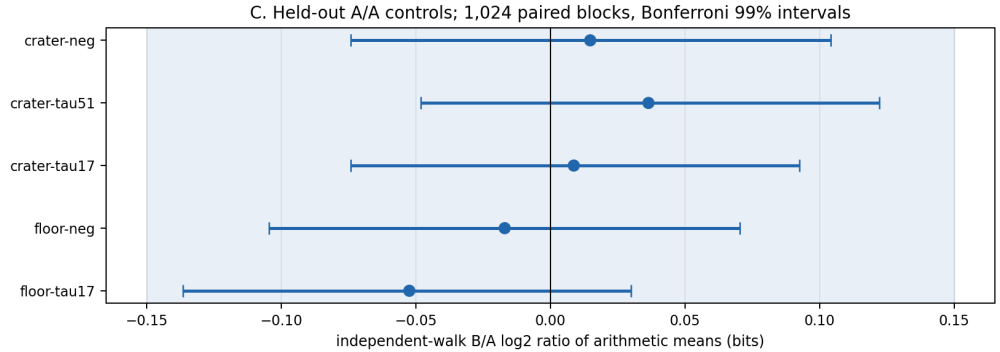
The ordinary-rho control was `floor-neg / crater-neg = -0.019008` bit. Its pointwise 95% CI `[-0.055582, +0.018144]` and Bonferroni 98.75% family interval `[-0.065653, +0.028718]` both include zero. The four action-order comparisons were compatible with their frozen random-map predictions: every predicted value lies inside its applicable family interval. Compatibility is not proof of that model.

The tested prime conductor step is real structural metadata: `f_E` changes from 1 to 271. Here `f_pi=271*120871`, but this run covers only the degree-271 / `f_E=271` slice. The factor-120871 descent and the bottom level were neither constructed nor measured. These results did not resolve a same-action walk-count difference at the frozen precision, but they do not identify conductor as a causal variable, establish conductor-independence, or price a degree-271 transfer.

m=51 conductor-271 crater/floor collision-walk record
Same-action association and action-order controls; not a causal conductor effect







Panel A: primary/diagnostic 95%; four secondary contrasts Bonferroni 98.75%. Panel B has no intervals. Panel C: Bonferroni 99%.

Study isogeny-volcano-m51-rho-20261013 | raw records SHA-256 0f3099774d453053... | hierarchical orbit-resampling intervals are sensitivity analyses

Figure source: plot_publication.py; vector rendering: volcano_level_contrasts.svg. Panel A shows fixed-exact-sample intervals: 95% for the primary and diagnostic, and Bonferroni 98.75% for the four-member secondary family. Panel B points are descriptive and have no frozen multiplicity-adjusted intervals. Panel C shows fixed-exact-sample Bonferroni 99% A/A intervals. The hierarchical orbit-resampling intervals are sensitivities and are not drawn. The separate footer band fixes the x-label/footer overlap in the frozen presentation plot.

Question and evidence status

Requested distinction	Evidence in this run	Status
Same action at both levels	`floor-tau17 / crater-tau17`	complete for the exact sample
Ordinary-rho level control	`floor-neg / crater-neg`	complete for the exact sample
Symmetry/action-order controls	three secondary quotient contrasts plus one diagnostic	complete at the `walk_ops` level
Floor-orbit coverage	one fixed representative from all 16 size-17 Frobenius orbits	complete
Randomisation/harness drift	five held-out independent-walk A/A contrasts	all frozen gates pass
Explicit vertical map and dual	not exercised	not assessed
Fully charged endpoint-to-endpoint or endpoint-to-crater cost	not measured	not assessed
Larger-subgroup or NIST scaling	no scaling ladder in this study	not assessed
Weak-curve claim	forbidden by the protocol and unsupported by these data	not assessed

Exact objects and workload

The frozen objects and design are in PROTOCOL.md, spec.json, and sample.csv. Work is over

`GF(2)[z] / (z^51 + z^6 + z^3 + z + 1)`

on `E_b: y^2 + x*y = x^3 + b`. The common ambient order is `2,251,799,852,369,764`; the tested prime subgroup has order `r=17,192,986,687` (`log2(r)=34.001101...`) and cofactor `130,972`. The Frobenius data are

`Delta_pi = -7*(271*120871)^2`, `D_K=-7`, and `f_pi=32,756,041`.

Only the crater and the `f_E=271` slice were constructed for this run. It did not construct or measure the separate factor-120871 descent or the bottom `f_E=f_pi` level, so it is not a measurement of the complete conductor gap.

sampled level	model inventory	`f_E`	`[End(E):Z[pi]] = f_pi/f_E`	native tested actions
crater, `b=1`	1	1	32,756,041	negation, `tau`, `tau^17`
conductor-271 floor	272 curves in 16 size-17 orbits; 16 fixed representatives tested	271	120,871	negation, `tau^17`

`tau` and `tau^17` are inseparable endomorphism actions, not separable volcano edges. Every sampled floor coefficient is in `GF(2^17)`, and the crater is defined over `GF(2)`. The order-`r` line therefore has the same degree-3 trace-zero context at both levels; it is not a discovered floor advantage. The algebraic background and still-open transfer obligations are recorded in TRACE_ZERO_LANE.md.

The 4,096 A-stream target blocks pair one planted scalar across all five arms. Within each floor orbit, offsets `64..255` supply 192 inferential blocks, for 3,072 total. Disjoint offsets `0..63` supply 64 A/A blocks per orbit, or 1,024 total. A separate B walk stream repeats those five held-out arms. Thus the sealed workload contains `20,480` A-stream plus `5,120` B-stream records: 25,600 verified solves.

Frozen contrasts

Displayed values are rounded to six decimals; the source decimal strings are preserved in contrast_summary.csv and analysis.json. The estimand is always `log2(arithmetic mean numerator walk_ops / arithmetic mean denominator walk_ops)`, not a mean of per-target ratios.

Primary same-action and ordinary-rho level controls

contrast	numerator mean	denominator mean	observed bits	fixed-sample interval	hierarchical sensitivity
primary `floor-tau17 / crater-tau17`	69,896.649414	69,618.020508	+0.005763	95% `[-0.032170, +0.043374]`	95% `[-0.045718, +0.059596]`
secondary `floor-neg / crater-neg`	121,282.560872	122,891.046875	-0.019008	Bonferroni 98.75% `[-0.065653, +0.028718]`	Bonferroni 98.75% `[-0.084160, +0.045691]`

For the same-negation row, the pointwise 95% interval is `[-0.055582, +0.018144]`. The primary fixed-sample interval retains all 16 sampled orbits exactly once; the hierarchical interval instead models an orbit superpopulation and is a sensitivity only. Both fixed-sample intervals include zero; the primary 95% interval is wholly inside `+/-0.10` bit.

Symmetry/action-order controls

ratio	observed bits	frozen prediction	residual	applicable family interval for observed	applicable interval for residual
`floor-tau17 / crater-tau51`	+2.030601	+2.043731	-0.013131	98.75% `[+1.982463, +2.078609]`	`[-0.061269, +0.034878]`
`floor-tau17 / floor-neg`	-0.795077	-0.792481	-0.002596	98.75% `[-0.841369, -0.748567]`	`[-0.048888, +0.043914]`
`crater-tau51 / crater-neg`	-2.844685	-2.836213	-0.008473	98.75% `[-2.891296, -2.797127]`	`[-0.055083, +0.039086]`
diagnostic `crater-tau17 / crater-neg`	-0.819847	-0.792481	-0.027366	95% `[-0.857563, -0.781709]`	`[-0.065082, +0.010772]`

The corresponding hierarchical sensitivity family intervals for the observed effects are, in row order, `[+1.971980, +2.089549]`, `[-0.851518, -0.738876]`, `[-2.912367, -2.776217]`, and `[-0.881503, -0.757131]`. The action-order rows deliberately change the signed action. They measure implementation-level symmetry effects, not level effects or transported actions.

Held-out A/A controls

All five primary A/A intervals lie inside the frozen `+/-0.15`-bit gate, so the inferential labels were released. These gates use the fixed 16-orbit exact sample. The hierarchical intervals are sensitivity analyses, not gate criteria; the `crater-tau51` upper bound and `floor-tau17` lower bound extend slightly beyond `+/-0.15`.

arm	A mean	B mean	B/A bits	fixed-sample Bonferroni 99% CI	hierarchical 99% sensitivity	gate
`crater-neg`	119,542.652344	120,775.493164	+0.014802	`[-0.074259, +0.104131]`	`[-0.096957, +0.130364]`	pass
`crater-tau51`	16,815.760742	17,245.871094	+0.036437	`[-0.048191, +0.122118]`	`[-0.083065, +0.150374]`	pass
`crater-tau17`	69,256.466797	69,675.470703	+0.008702	`[-0.074166, +0.092426]`	`[-0.105748, +0.125577]`	pass
`floor-neg`	120,694.543945	119,283.780273	-0.016963	`[-0.104524, +0.070310]`	`[-0.147229, +0.108921]`	pass
`floor-tau17`	70,057.581055	67,559.016602	-0.052393	`[-0.136468, +0.030002]`	`[-0.164949, +0.058201]`	pass

Per-orbit observations

Each point below uses the 192 inferential blocks attached to the named fixed floor representative. These are descriptive/exploratory points: the frozen protocol supplies no per-orbit confidence intervals or multiplicity adjustment.

orbit	exact `b`	primary `floor-tau17 / crater-tau51` bits
1	623162416261	+0.1507931966
2	25640824825979	-0.0252056902
3	30885819784827	-0.0263654466
4	57318330399736	+0.0106200590
5	67694307705004	-0.0405102320
6	72881269935803	+0.1655014433
7	115635957935040	-0.0637974688
8	116461223646592	+0.0213458470
9	116511665573478	-0.0377513252
10	121360481466206	-0.0197344347
11	130246110195364	-0.0041532025
12	147686731444433	+0.1054013366
13	213718369432541	-0.1157022526
14	237711860122495	+0.0352762381
15	360860333097189	-0.0928367359
16	370376419722289	+0.0417179550

The exact ranges for all six contrasts are:

contrast	minimum (orbit, `b`)	maximum (orbit, `b`)
primary `floor-tau17 / crater-tau17`	-0.115702 (13, 213718369432541)	+0.165501 (6, 72881269935803)

contrast	minimum (orbit, b)	maximum (orbit, b)
floor-neg / crater-neg	-0.171881 (15, 360860333097189)	+0.105389 (13, 213718369432541)
floor-tau17 / crater-tau51	+1.941811 (3, 30885819784827)	+2.125624 (6, 72881269935803)
floor-tau17 / floor-neg	-0.877937 (10, 121360481466206)	-0.708363 (5, 67694307705004)
crater-tau51 / crater-neg	-2.950169 (10, 121360481466206)	-2.716883 (13, 213718369432541)
diagnostic crater-tau17 / crater-neg	-0.996209 (1, 623162416261)	-0.594998 (13, 213718369432541)

All 96 exact orbit-by-contrast rows, their means, predictions, and residuals are preserved in orbit_summary.csv; all 80 orbit-by-arm distribution rows are in orbit_arm_summary.csv.

Operation accounting

The primary metric, walk_ops, counts online group additions and doublings from collision-search start through successful and abandoned walks, including look-ahead work, starts, and cycle escapes. Every record has 162 separately counted target-dependent setup operations: 34 doublings plus 128 additions. mean total ops = mean walk_ops + 162 exactly. Quantiles are across the 3,072 inferential records per arm.

arm	mean walk ops	SD	q05	median	q95	mean total ops
crater-neg	122,891.046875	63,931.153002	30,241.850000	115,439.500000	237,425.550000	123,053.046875
crater-tau51	17,107.424479	8,863.338054	4,633.650000	15,960.000000	33,126.800000	17,269.424479
crater-tau17	69,618.020508	36,571.426218	17,258.700000	64,689.000000	137,298.600000	69,780.020508
floor-neg	121,282.560872	62,787.958935	34,153.700000	112,891.000000	232,938.400000	121,444.560872
floor-tau17	69,896.649414	35,887.774600	18,779.100000	65,973.000000	135,324.000000	70,058.649414

Canonicalisation primitives are recorded, not silently treated as free. No field-primitive weights were frozen, so they are not converted to group-addition equivalents.

arm	mean canonicalisations	mean x-squares	mean y-squares	mean sign tests	mean negations
crater-neg	119,196.759115	0	0	119,196.759115	59,600.025065
crater-tau51	16,593.013021	829,650.651042	414,769.905924	16,593.013021	8,294.725586
crater-tau17	67,524.067057	2,295,818.279948	1,147,894.264974	67,524.067057	33,762.790039
floor-neg	117,635.792969	0	0	117,635.792969	58,823.859375
floor-tau17	67,794.073568	2,304,998.501302	1,152,477.198242	67,794.073568	33,900.784505

arm	mean walk starts	mean cycles	mean useless collisions	mean seconds (descriptive)
crater-neg	3,694.287760	2.107747	0.024089	0.088924
crater-tau51	514.411458	0.003906	0	0.026773
crater-tau17	2,093.953451	0.186523	0.004883	0.088962
floor-neg	3,646.767904	2.078451	0.020508	0.087755
floor-tau17	2,102.575846	0.182943	0.006510	0.089213

Mean abandoned walks were zero in every arm. Seconds are descriptive because the host was not claimed to be isolated. Full distributions and fixed- and hierarchical-sample mean intervals are in arm_summary.csv.

The accounting interval excludes one-time generator/Ust construction and certificates, planted-target construction, final recovery multiplication and certificate verification, and weights for canonicaliser field primitives. Consequently neither walk_ops nor total counted group operations is a fully charged end-to-end attack cost.

Statistical method

The frozen analyzer used 50,000 paired percentile-bootstrap draws with root seed 202610135106.

- The governing exact-sample bootstrap retains each of the 16 sampled orbit strata once and resamples the 192 paired inferential blocks within each.
- The separately reported hierarchical sensitivity resamples orbit strata and then paired blocks. It targets a modeled orbit superpopulation, not the exact sampled-orbit census.
- Primary and diagnostic intervals are pointwise 95%. The four secondary rows use Bonferroni 98.75% marginal intervals for 95% family coverage.

- The five A/A gates use disjoint held-out data and Bonferroni 99% marginal intervals for 95% family coverage.
- The primary equivalence decision requires its complete 95% interval to lie inside `+/-0.10` bit. Containing zero and establishing equivalence are separate questions.

The one chosen crater, fixed floor representatives, generator construction, hash behavior, and implementation are bundled with level. Pairing reduces target noise; it does not randomize conductor or identify a causal conductor effect.

Integrity and provenance

The run started at `2026-10-13T07:49:46.351340+00:00` and finished at `2026-10-13T07:58:34.011824+00:00`; the 400-job interval was `521.079691130999` seconds. Four workers were pinned to cores 0--3 on an AMD EPYC 9V74. The clean producer revision before and after the frozen snapshot was `929ccbcfce51d2652ecc5f647f15660a988c9aba`. The compiler was GCC 14.2.0. The exact invocation, host record, receipts, raw outputs, and seal are inside `run1.tar.zst`; replay instructions and independent audit notes are in `ARCHIVE.md`.

artifact	SHA-256
frozen-input commitment `FROZEN.sha256`	`149a9028aeda673dd833fb699543b9c7f6d7ed59dff17e5f1d18d96a505508d9`
complete archive `run1.tar.zst`	`eaad213c09f70c4c3cc330af20286328e51ab079545abaf5b80feba1b0027505`
sealed-tree `MANIFEST.sha256` inside archive	`1318653f04d8c1a8b79d6cd646d0f1a410fe78c5e13c9db2d1700006b7a04d1d`
normalized raw records inside archive	`0f3099774d4530538763723a735e2244586ab40bfb8589dda9c32cc6413705f8`
solver binary inside archive	`4db25d2113bc39cce71e24b317b4c3ded69a7e177a23f86c4dbb2277bafdb3e0`
`analysis.json`	`9087c623b85ab0da7dc483deb7d3153eeae1ded8f87c5283b912d36c5e7feee6`
`arm_summary.csv`	`1dfd292827c456e1ddbefeddd13b69cdaabbfe98705cd85d86aa1daae573470b`
`contrast_summary.csv`	`4bae750a075b408c02d942fd81e65b176926f34cd53d78dd119d4ee10066ba96`
`orbit_summary.csv`	`e1c29093a1e8523badb901c45698d52844b58538297038066ae0738ed555df51`
`orbit_arm_summary.csv`	`3aed1593abea428fc4dba87d02f25c78c0d343ceae34a28821a2d29f62c344e6`
`aa_summary.csv`	`3bcd30cf1178edb8ab04ef21a93d5a2389f7b2b40d0b1adc494a35edeb646d29`
publication plot source `plot_publication.py`	`6de756bc56e57db179cf3ab5379b7bce51b9b88c0e411cef3d941db365cff388`
publication SVG	`246c602e3666042243563da36a8fee5e47acfd9f27cfc440071523cd42b41252`
publication PNG	`e85d7fe220173f316227af6b0bb3dbbd1d5a0e4b2dd882ed0946a60297db2ea7`

The archive holds 400 raw files, 400 job receipts, 35 self-test certificates, and a 424-entry file seal. The analyzer independently rechecked the frozen commitment, exact file sets, plan, commands, receipt bindings, raw rows, counter identities, scalars, stream namespaces, orbit assignment, binary, session, and complete seal before issuing `VALID_FOR_SCOPED_INFERENCE`.

Reproduce without changing the evidence record

From the repository root, extract to a new empty directory, verify the sealed tree, and direct reanalysis to a separate new output directory:

```
study_dir=research/isogeny_volcano_m51_rho_20261013
replay_root=/tmp/isogeny-volcano-m51-rho-replay
replay_out=/tmp/isogeny-volcano-m51-analysis-replay
test ! -e "$replay_root"
test ! -e "$replay_out"
mkdir "$replay_root"
tar --zstd -xf "$study_dir/run1.tar.zst" -C "$replay_root"
(cd "$replay_root" && sha256sum --strict -c MANIFEST.sha256)
python3 "$study_dir/analyze.py" \
  --run-root "$replay_root" \
  --output-dir "$replay_out"
MPLCONFIGDIR=/tmp/m51-volcano-replay-mpl \
  python3 "$study_dir/plot_publication.py" \
  --input-dir "$replay_out" \
  --output-prefix /tmp/m51-volcano-replay
```

Build a byte-reproducible landscape PDF with the installed Pandoc/pdfTeX toolchain. The trailer ID is derived from the report source rather than time:

```
report_source=research/isogeny_volcano_m51_rho_20261013/README.md
report_sha256=$(sha256sum "$report_source" | awk '{print $1}')
report_id=${report_sha256:0:32}
SOURCE_DATE_EPOCH=1791849600 \
FORCE_SOURCE_DATE=1 \
```

```
pandoc "$report_source" \
  --from=gfm --standalone --pdf-engine=pdflatex \
  --resource-path=research/isogeny_volcano_m51_rho_20261013 \
  -V documentclass=extarticle -V fontsize=8pt \
  -V geometry:landscape -V geometry:margin=0.45in -V colorlinks=true \
  -V "header-includes=\\pdftrailerid{<${report_id}><${report_id}>}" \
  -o research/isogeny_volcano_m51_rho_20261013/report.pdf
```

Claim boundary

This record supports only bounded statements about collision-walk operation counts for this prime subgroup, implementation, exact crater, fixed set of 16 floor representatives, generator construction, hash behavior, and accounting interval. It does **not** establish:

- a causal effect of the conductor or the degree-271 vertical edge;
- an explicit isogeny/dual, subgroup-preserving DLP transfer, or recovery path;
- fully priced canonicalisation, map evaluation, one-time transfer, or end-to-end attack cost;
- scaling to larger subgroups or any NIST P-, B-, K-, or sect113r1 curve;
- a security-bit reduction or a weak curve.

The next transfer claim would require explicit maps and duals, subgroup and scalar-recovery certificates, and complete cold-start and amortised cost against the best same-subgroup control. These data do not substitute for any of those gates.